F 13

Sinusformade växelströmmen

Komplexa metoden

2025-01-20

S. 226 - 243

+ Bild 9 - 11 från F10

Fysiska (mätbara) spänningar och strömmar:

$$u(t) = \hat{u} \cdot \cos(\omega t + \alpha) \qquad i(t) = \hat{i} \cdot \cos(\omega t + \beta)$$

Komplex form:

"Analytisk spänning/ström"

$$\bar{u}(t) = \hat{u} e^{j(\omega t + \alpha)} \qquad \bar{i}(t) = \hat{i} e^{j(\omega t + \beta)}$$

$$u(t) = \mathrm{Re}\{\bar{u}(t)\} \qquad i(t) = \mathrm{Re}\{\bar{i}(t)\}$$

$$\bar{u}(t) = \underbrace{\hat{u} e^{j\alpha}}_{\bar{U}} \cdot e^{j\omega t} \qquad \bar{i}(t) = \underbrace{\hat{i} e^{j\beta}}_{\bar{I}} e^{j\omega t}$$

Skrivs $\bar{U}$

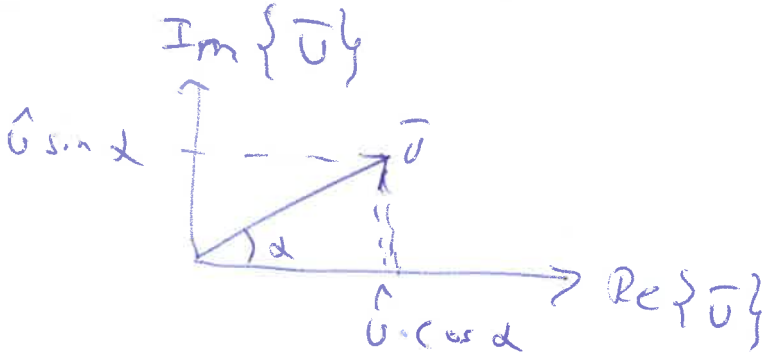
Skrivs $\bar{u}$ ← komplex vidare

$$\bar{U} = \hat{u} e^{j\alpha} = \hat{u} \angle \alpha$$

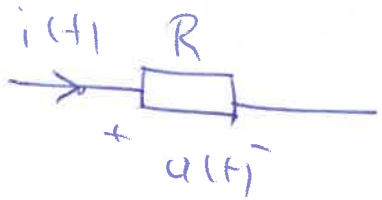
Finns med som gemensam faktor i alla uttryck
⇒ slopas ofta med

$\bar{U} = \hat{U} e^{j\alpha}$ ← polär form

$= (\hat{U}\cos\alpha, \hat{U}\sin\alpha)$ på rektangulär form:



Resistor:



$u(t) = R \cdot i(t) =$

$= R \cdot \hat{i} \cos(\omega t + \beta) =$

$= \mathrm{Re}\{R \cdot \hat{i} \cdot e^{j(\omega t + \beta)}\}$

$\Rightarrow \bar{U} = \underbrace{R \cdot \hat{i}}_{\hat{U}} e^{j\beta} = R\bar{I}$ ($\beta = \alpha$)

$$\boxed{\bar{U} = R \cdot \bar{I}}$$

$\bar{U}$ i fas med $\bar{I}$

Spole:



$u(t) = L \cdot \frac{di}{dt}$

$= -L \cdot \hat{i} \cdot \omega \sin(\omega t + \beta) =$

$= \omega L \hat{i} \sin(-(\omega t + \beta)) =$

$= \omega L \hat{i} \cos(\omega t + \beta + 90^\circ) =$ ($+90^\circ$ ← $-\sin$)

$= \mathrm{Re}\{\omega L \hat{i} \cdot e^{j(\omega t + \beta + 90^\circ)}\}$

$\Rightarrow \bar{U} = \underbrace{\omega L \hat{i}}_{\hat{U}} e^{j(\beta + 90^\circ)} = \alpha$

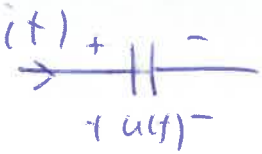
$= \omega L \hat{i} \underbrace{e^{j90^\circ}}_{j} \cdot e^{j\beta} =$

$= j\underbrace{\omega L \hat{i}}_{\hat{U}} e^{j\beta}$

$$\Rightarrow \boxed{\bar{U} = j\omega L \cdot \bar{I}}$$

$\bar{U}$ 90° före $\bar{I}$

Kondensator:



$i = C \cdot \frac{du}{dt}$

$u(t) = \frac{1}{C} \cdot \int i(t)\,dt$

$= +\frac{1}{C} \cdot \frac{\hat{i}}{\omega} \sin(\omega t + \beta) =$

$= \frac{\hat{i}}{\omega C} \cos(\omega t + \beta - 90^\circ)$ ($\hat{i}$ ← -90)

$= \mathrm{Re}\{\underbrace{\frac{\hat{i}}{\omega C}}_{\hat{U}} \cdot e^{j(\omega t + \underbrace{\beta - 90^\circ}_{\alpha})}\}$

$\Rightarrow \bar{U} = \frac{\hat{i}}{\omega C} \underbrace{e^{-j90^\circ}}_{-j = \frac{1}{j}} \cdot e^{j\beta}$

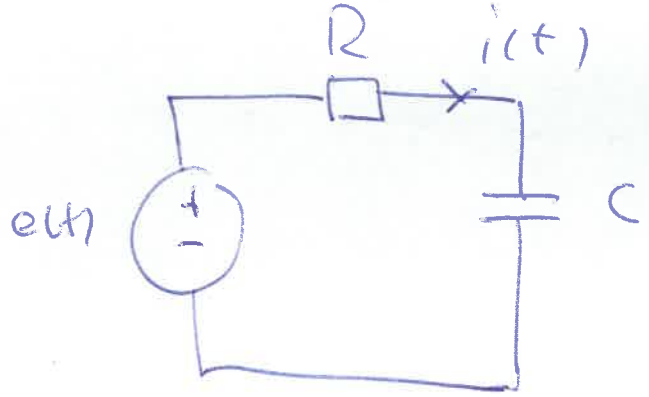
$= -j \cdot \frac{1}{\omega C} \cdot \bar{I} = \frac{1}{j\omega C} \cdot \bar{I}$

$$\boxed{\bar{U} = -j\frac{\bar{I}}{\omega C} = \frac{1}{j\omega C}\bar{I}}$$

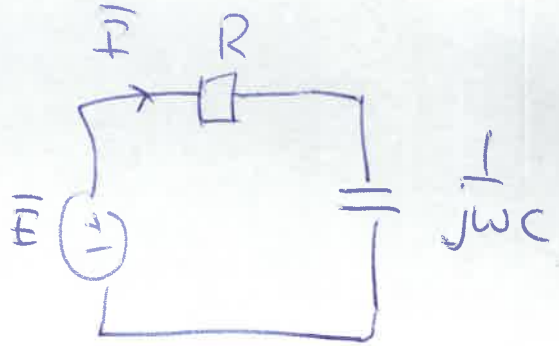
$\bar{U}$ 90° efter $\bar{I}$

(4)

Ex: 9.15



Komplex "phasor" domän



$R = 100\,\Omega$

$C = 5\,\mu F$

$\omega = 1000$ rad/s

$e(t) = 10 \cdot \cos(1000t) \Rightarrow \bar{E} = 10\angle 0°$

$\uparrow \updownarrow$ $\uparrow \omega t$ $\uparrow \alpha = 0°$

$$\bar{I} = \frac{\bar{E}}{R + \frac{1}{j\omega C}} = \frac{j\omega C}{1 + j\omega RC} \cdot \bar{E}$$

$\bar{Z} = R - j\cdot\frac{1}{\omega C}$

$|\bar{Z}| = \sqrt{R^2 + (1/\omega C)^2}$

$\arg \bar{Z} = \arctan\left(\frac{-1/\omega C}{R}\right) =$

$-\arctan\left(\frac{1}{\omega RC}\right)$

$$\frac{\bar{E}}{R - j\cdot\frac{1}{\omega C}} = \frac{E\angle 0°}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}\,\angle -\arctan\frac{1}{\omega RC}}$$

$\sqrt{100^2 + \frac{1}{(1000\cdot 5\cdot 10^{-6})^2}} =$

$\sqrt{100^2 + 200^2} = 100\cdot\sqrt{5} = 223{,}6$

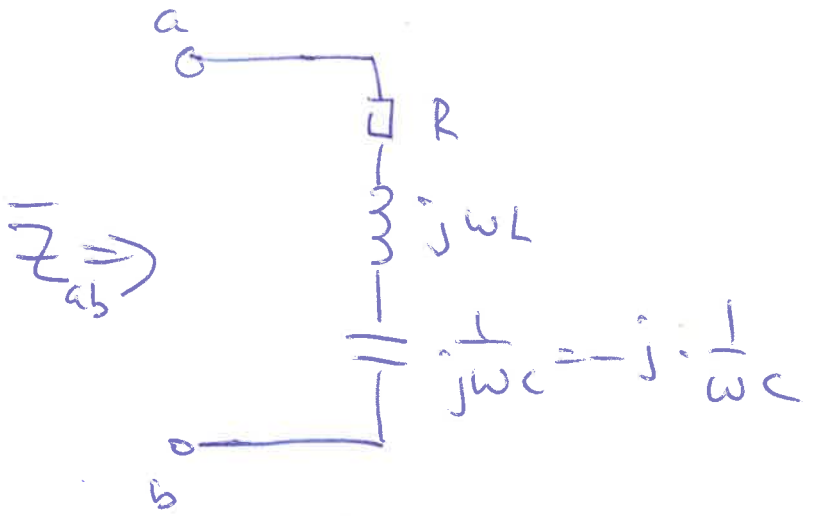
$$\frac{10\angle 0°}{223{,}6\angle -63{,}43°} =$$

$= 44{,}7\angle 63{,}43°$ mA

$\arctan\left(\frac{1}{\omega RC}\right) = \arctan\left(\frac{1}{1000\cdot 100\cdot 5\cdot 10^{-6}}\right) = \arctan(2) = 63{,}43°$

Komplex impedans, impedansvinkel, impedansdiagram:

Ex: Serie-krets



$$\bar{Z}_{ab} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$Z_{ab} = |\bar{Z}_{ab}| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

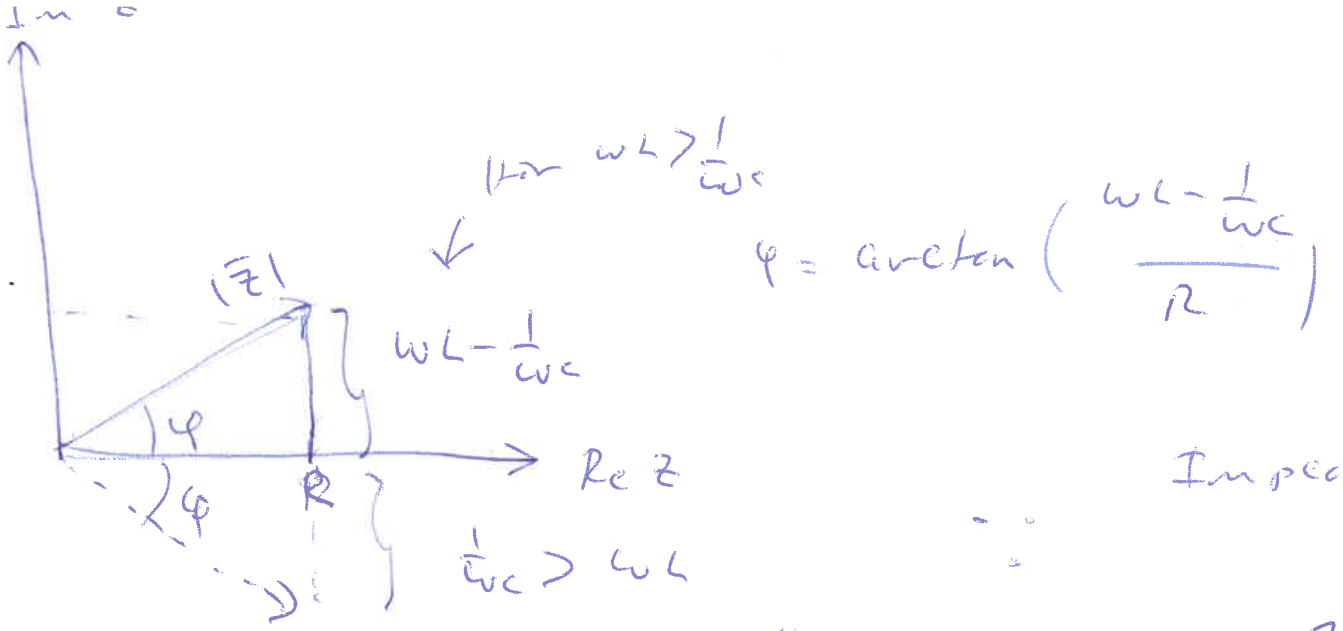
$$\angle \bar{Z}_{ab} = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

$\uparrow$

$\arg\{\bar{Z}_{ab}\}$

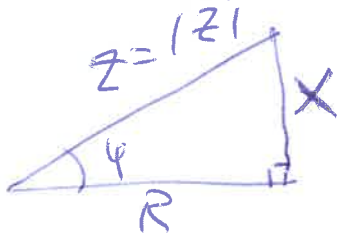
Specialfall: $\omega L = \frac{1}{\omega C} \Rightarrow \mathrm{Im}\{\bar{Z}_{ab}\} = 0$ $\Rightarrow \bar{Z}_{ab} = R$

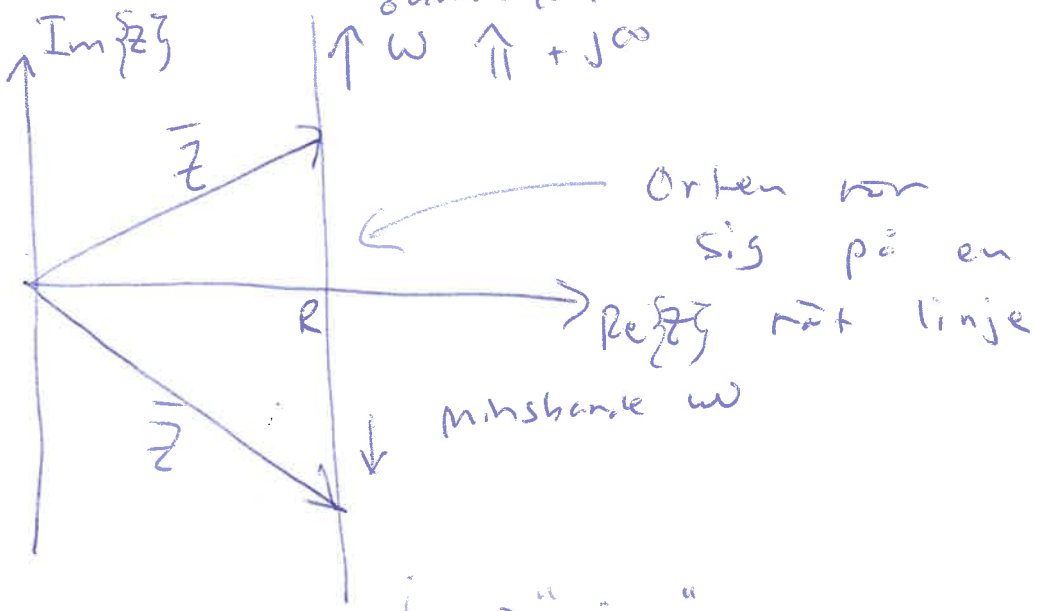
D: $\omega = \frac{1}{\sqrt{LC}}$ $f = \frac{1}{2\pi\sqrt{LC}}$ Resonansfrekvens



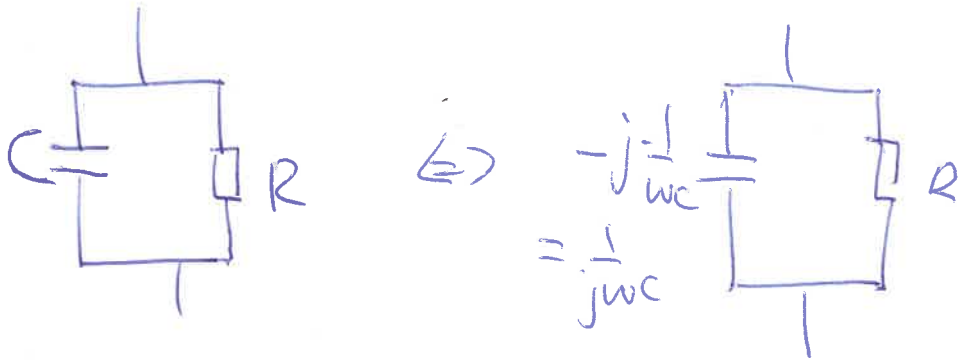
$$\varphi = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

Impedans triangel.





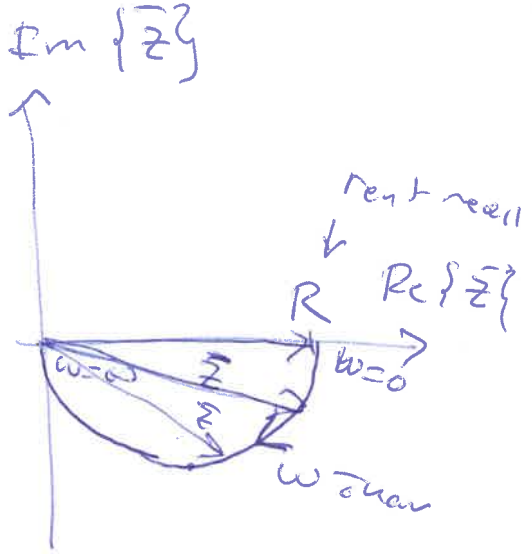
Ex:



$$\bar{Z} = \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{R}{1 + j\omega RC}$$

adim. $\omega = \frac{rad}{s}$ $RC = s$

$$|\bar{Z}| = \frac{R}{\sqrt{1+(\omega RC)^2}}$$



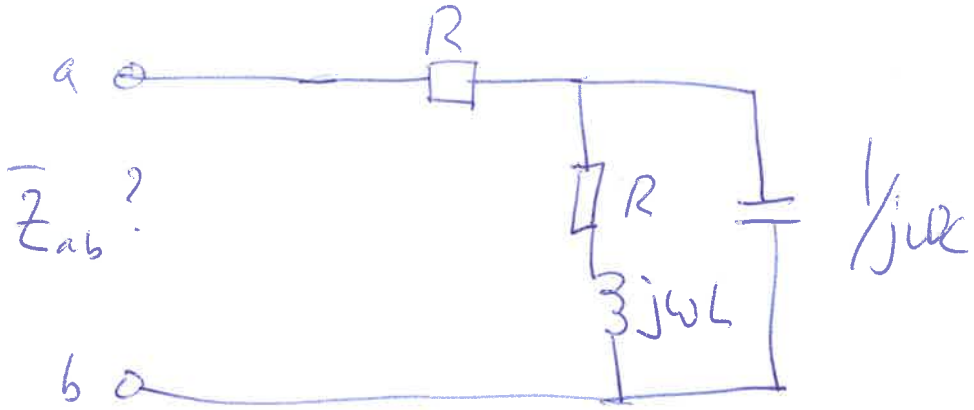
$$\bar{Z} = \frac{R \cdot (1 - j\omega RC)}{\cancel{\cdot}(1+(\omega RC)^2)} = \frac{R}{(1+\omega RC)^2} - \frac{j\omega RC \cdot R}{1+(\omega RC)^2}$$

← 0 dacă $\omega = 0$

← 0 dacă $\omega \to \infty$

Im - cel mai mic dacă $\omega RC = 1 \Rightarrow \text{Im}\{\bar{Z}\} = \frac{R}{2}$

Impedansräkning p.s.s som för Resistorer



Obs! $\bar{Z} = j\omega L = jX_L$

$\bar{Z} = \frac{1}{j\omega C} = -j\cdot\frac{1}{\omega C} = -j\cdot X_C$

$$\bar{Z}_{ab} = R + \frac{(R+j\omega L)\cdot\frac{1}{j\omega C}}{R+j\omega L+\frac{1}{j\omega C}} =$$

$$= R + \frac{-j\frac{1}{\omega C}\cdot(R+j\omega L)}{R+j(\omega L-\frac{1}{\omega C})} = \left\{ \begin{array}{l} \text{Förlora} \\ \text{med} \\ \text{konjugatet} \end{array} \right\}$$

$$= R - j\cdot\frac{1}{\omega C}\cdot\frac{(R+j\omega L)\cdot(R-j(\omega L-\frac{1}{\omega C}))}{R^2+(\omega L-\frac{1}{\omega C})^2} = \text{forts.}$$

forts:

$$\bar{Z}_{ab} = R - j\cdot\frac{1}{\omega C}\cdot\frac{R^2 + \omega L\left(\omega L - \frac{1}{\omega C}\right) + j\left(\omega LR - R\left(\omega L - \frac{1}{\omega C}\right)\right)}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$= R - j\cdot\frac{1}{\omega C}\cdot\left(\frac{R^2 + \omega L\left(\omega L - \frac{1}{\omega C}\right) + j\frac{R}{\omega C}}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}\right) =$$

$$R - j\cdot\frac{1}{\omega C}\cdot\left(\frac{1 + \omega\frac{L}{R}\cdot\left(\omega\frac{L}{R} - \frac{1}{\omega RC}\right) + \frac{j}{\omega RC}}{1 + \left(\frac{\omega L}{R} - \frac{1}{\omega RC}\right)^2}\right)$$

Svårt att snygga till

~~$\omega \to 0$~~ $\to$ $R - j\frac{1}{\omega C}\left(\frac{1 + \frac{L}{R^2C} + \frac{1}{\omega RC}}{1 + \left(\frac{1}{\omega RC}\right)^2}\right)$

Asymptotische Fall:

$$\omega \to 0 \Rightarrow \underline{Z}_{ab} \to R + \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = R + \frac{R}{1 + j\omega RC} \to 2R$$

$$\omega \to \infty \quad \underline{Z}_{ab} \to R + \frac{1}{j\omega C} \to R$$